Lab 4: Fitting a GEV to a tide gauge

2026-09-18

In this lab, you will fit a generalized extreme value (GEV) distribution to annual maximum water levels at a tide gauge by maximum likelihood, by L-moments, and by Bayesian inference, then compare the estimates.

The code below runs end to end on Sewells Point, Virginia. Then you change one line to select a different gauge and rerun it. A different gauge may need different detrending and prior choices.

Objectives

By the end of this lab, you will be able to:

1. Fit a GEV by maximum likelihood and by L-moments, and say why the two disagree.
2. Write a GEV log-likelihood from the density and optimize it yourself.
3. Read a prior predictive simulation and say whether the prior is defensible.
4. Diagnose a Markov chain that has not converged, and say what to do about it.

Setup

The cell below activates this lab's own package environment, installs what it lists, and loads the packages. Every lab starts with the same cell. It also creates the random number generator that the rest of the lab draws from, seeded so a fresh render gives the same figures every time.

Four of these packages are new this week.

- `CEVE543Utils` wraps GEV fitting and NOAA data loading for this course
- `Turing` writes Bayesian models and draws from their posteriors (documentation)
- `Optim` minimizes the likelihood you write by hand
- `Unitful` attaches physical units to numbers

```
using Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()

using CEVE543Utils
using CairoMakie, DataFrames, Distributions, LaTeXStrings
using Optim, Printf, Random, Statistics, Turing, Unitful
using FlexiChains: summarystats

rng = MersenneTwister(543)
set_theme!(theme_minimal(); fontsize=14)
colors = Makie.wong_colors()
```

Line 8

Turing stores its output in a `FlexiChain`, and `summarystats` is the function that turns one into the table of convergence diagnostics used at the end of this lab.

Line 10

Passed to every function below that draws at random, so the results are reproducible.

Lab activity

The data

`load_water_level` downloads every hourly reading a NOAA tide gauge has published and caches it beside this notebook. Sewells Point sits in Norfolk, Virginia, at the mouth of Chesapeake Bay, and has reported since 1928.

```
station = "8638610"
record = load_water_level(
    station;
    name="Sewells Point",
    cache=joinpath(@__DIR__, "$station-hourly.csv"),
)
```

Line 1

The line you will change later. Station ids are listed at NOAA's station browser; pick one with a long record.

Line 5

Where the download is kept. `@__DIR__` is this notebook's own folder, so the cache is found whether you render the notebook or run cells in the REPL. The first download of a century-long record takes a minute or two; every run after that reads the cache in seconds.

```
WaterLevelRecord: 851121 readings in m
  station: Sewells Point (8638610), MSL datum
  period:  1928-01-01T00:00:00 to 2026-10-31T23:00:00
  levels:  -1.504 to 2.032 m
```

The levels come back with units attached. `Unitful` attaches units to numbers, so a length cannot be added to a time and unit conversion is a function call rather than a hardcoded factor.

```
reading = record[1]
@printf("%s at %s\n", reading.level, reading.time)
@printf("in feet: %s\n", uconvert(u"ft", reading.level))
@printf("as a bare number: %.3f\n", ustrip(reading.level))
```

Line 3

`uconvert` changes the unit and the number together, so the physical quantity is unchanged.

Line 4

`ustrip` throws the unit away and hands back the number. Anything that does not understand units, which includes every fitting function in this lab, needs this.

```
-0.468 m at 1928-01-01T00:00:00
in feet: -1.535433070866142 ft
as a bare number: -0.468
```

Annual maxima

```
annmax = AnnMaxRecord(record; detrend=:msl, units=u"ft")
```

Line 1

`:msl` subtracts each year's mean sea level, leaving the water level above that year's mean. `:linear` subtracts an ordinary least squares fit to the maxima instead, and `:none` leaves the trend in. Try each one.

```
AnnMaxRecord: 97 years in ft, detrended by :msl
  station: Sewells Point (8638610), MSL datum
  period:  1928 to 2026
  levels:  3.5716335313204626 to 7.984411953308337 ft
```

An observed water level is the sum of mean sea level, the astronomical tide, storm surge, and vertical land motion. Sewells Point has both eustatic sea-level rise and subsidence, so its mean sea level rises about 4.6 mm/yr. A stationary GEV requires that every year be drawn from the same distribution, so the trend must be removed first. Subtracting annual mean sea level removes the sea-level component but leaves the tide in the residual. A design study would remove the tide as well.

```
y = ustrip.(waterlevels(annmax))
raw = y .+ annmax.baseline .- mean(annmax.baseline[(end-4):end])

fig = Figure(; size=(700, 320))
ax = Axis(fig[1, 1]; xlabel="year", ylabel="Annual Maximum [ft]")
scatter!(ax, obsyears(annmax), raw; color=(:gray, 0.6), label="observed")
scatter!(ax, obsyears(annmax), y; color=colors[1], label="detrended")
lines!(ax, obsyears(annmax), annmax.baseline; color=colors[2], label="mean sea level")
axislegend(ax; position=:lb)
fig
```

Line 1

The fitting functions below all want bare numbers, so this is the vector every fit uses.

Line 2

Undoing the detrending to show what came in. `AnnMaxRecord` keeps the baseline it subtracted for exactly this reason.

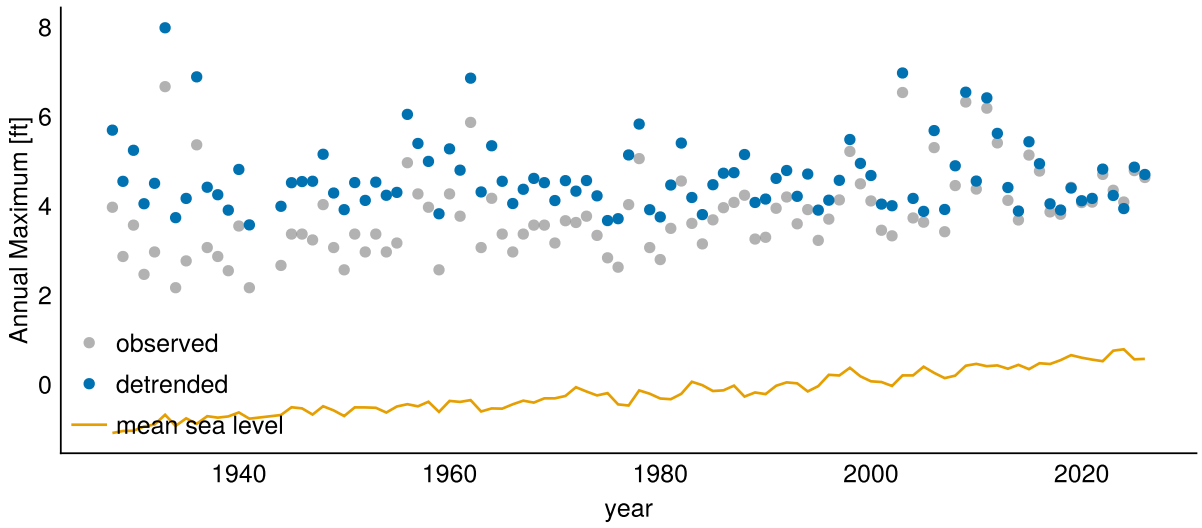


Figure 1: Annual maxima at Sewells Point before and after removing mean sea level. The baseline is what was subtracted.

Maximum likelihood and L-moments

`gevfit` fits by maximum likelihood and `gevfit_lmom` by L-moments. Both follow `Extremes.jl`.

```
fit_mle = only(getdistribution(gevfit(y)))
fit_lmom = gevfit_lmom(y)

@printf("MLE:       μ = %.3f, σ = %.3f, ξ = %.3f\n", fit_mle.μ, fit_mle.σ, fit_mle.ξ)
@printf("L-moments: μ = %.3f, σ = %.3f, ξ = %.3f\n", fit_lmom.μ, fit_lmom.σ, fit_lmom.ξ)
```

```
MLE:       μ = 4.258, σ = 0.484, ξ = 0.197
L-moments: μ = 4.259, σ = 0.492, ξ = 0.179
```

The GEV log-likelihood

You can also write the log-likelihood yourself. The log-density, with $v = 1 + \xi(y - \mu)/\sigma$, is:

$$\log f(y \mid \mu, \sigma, \xi) = -\log \sigma - \left(1 + \frac{1}{\xi}\right) \log v - v^{-1/\xi}.$$

Two cases need handling. When $v \le 0$ the observation is outside the support and the density is zero, so the log-density is $-\infty$. When $\xi = 0$ the expression divides by zero, but the distribution is the Gumbel there and its log-density is $-\log \sigma - z - e^{-z}$ with $z = (y - \mu)/\sigma$.

```
function my_gev_logpdf(y, μ, σ, ξ)
    σ <= 0 && return -Inf
    z = (y - μ) / σ
    abs(ξ) < 1e-7 && return -log(σ) - z - exp(-z)
    v = 1 + ξ * z
```

```
    v <= 0 && return -Inf
    return -log(σ) - (1 + 1 / ξ) * log(v) - v^(-1 / ξ)
end
```

Line 2

A negative scale is not a distribution. Returning `-Inf` instead of raising an error lets the optimizer treat the point as impossible and continue.

Line 4

The Gumbel limit. Testing `abs(ξ) < 1e-7` rather than `ξ == 0` catches the values an optimizer proposes near zero.

Line 6

The support condition. As μ, σ and ξ change, the set of observations with finite density changes.

Now minimize the negative log-likelihood. Optimize over $\log \sigma$ so the scale stays positive.

```
function negative_loglik(θ, y)
    μ, log_σ, ξ = θ
    return -sum(my_gev_logpdf(yᵢ, μ, exp(log_σ), ξ) for yᵢ in y)
end

θ₀ = [mean(y), log(std(y)), 0.1]
result = optimize(θ -> negative_loglik(θ, y), θ₀, NelderMead())
μ̂, log_σ̂, ξ̂ = Optim.minimizer(result)

@printf("converged: %s\n", Optim.converged(result))
@printf("by hand:   μ = %.3f, σ = %.3f, ξ = %.3f\n", μ̂, exp(log_σ̂), ξ̂)
@printf("gevfit:    μ = %.3f, σ = %.3f, ξ = %.3f\n", fit_mle.μ, fit_mle.σ, fit_mle.ξ)
```

Line 3

Summing the log-densities and negating, since `optimize` minimizes.

Line 6

The starting guess: sample moments for location and scale, and a small positive shape. A bad starting guess is the usual reason this fails.

Line 10

Check convergence before using the result.

```
converged: true
by hand:   μ = 4.258, σ = 0.484, ξ = 0.197
gevfit:    μ = 4.258, σ = 0.484, ξ = 0.197
```

> **!** Important
>
> If `converged` prints `false`, the parameters are not a maximum likelihood estimate and nothing downstream of them is valid. Try a different starting guess or a different algorithm.

Return levels from the three fits

Plot return level against return period, with the observations at Weibull plotting positions.

```
return_periods = exp.(range(log(1.1), log(500); length=60))
fit_hand = GeneralizedExtremeValue(μ̂, exp(log_σ̂), ξ̂)

sorted_obs, _, observed_periods = plotting_positions(y)

fig = Figure(; size=(700, 400))
ax = Axis(fig[1, 1]; ylabel=L"\text{return level (ft)}")
for (d, colour, style, label) in (
    (fit_mle, colors[1], :solid, "gevfit (MLE)"),
    (fit_lmom, colors[2], :solid, "L-moments"),
    (fit_hand, colors[3], :dash, "by hand"),
)
    levels = [quantile(d, 1 - 1 / T) for T in return_periods]
    lines!(
        ax, return_periods, levels;
        color=colour, linewidth=2, linestyle=style, label=label
    )
end
scatter!(ax, observed_periods, sorted_obs; color=:black, markersize=6,
label="observed")
return_period_axis!(ax)
axislegend(ax; position=:lt)
fig
```

Line 1

Sixty return periods from just over 1 year to 500, log-spaced so the short ones do not crowd together.

Line 2

Your hand-fitted parameters, packaged as a distribution so one `quantile` call works on all three.

Line 4

Sorts the record and pairs each observation with the return period its rank implies.

Line 11

Dashed because your fit and `gevfit` maximize the same likelihood, so the two curves coincide. The dashed line following the solid one confirms your log-density is correct.

Line 13

The T-year return level is the quantile with exceedance probability $1/T$.

Line 20

Log axis with return periods as tick labels.

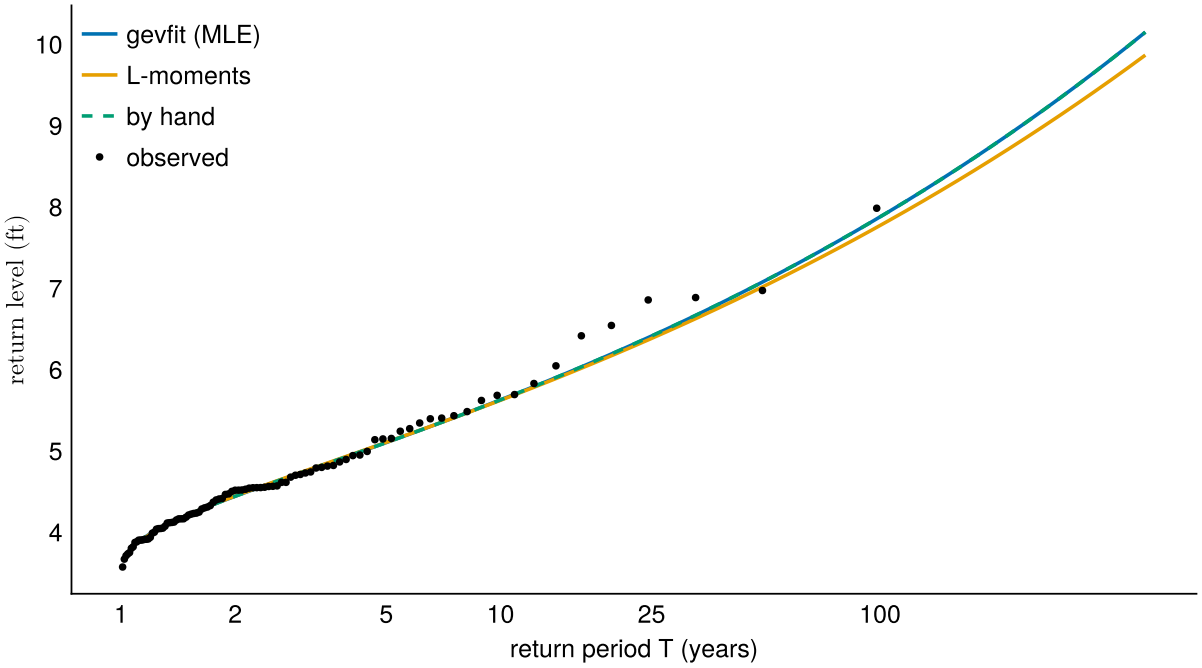


Figure 2: Return levels from the three fits, with the observed record at Weibull plotting positions.

Bootstrap confidence bands

Each of those curves is one estimate from 96 years of data. A different 96 years from the same process would give different parameters and a different curve. The bootstrap estimates that spread: resample the record with replacement, refit, and repeat.

Bootstrap the L-moment fit rather than the maximum likelihood fit. L-moments are closed form, so every resample returns an estimate. A maximum likelihood fit fails on some resamples, and the fits that survive are a biased subset.

```
n_boot = 1_000
boot_fits = [gevfit_lmom(y[rand(rng, 1:length(y), length(y))]) for _ in 1:n_boot]
boot_curves = [[quantile(d, 1 - 1 / T) for T in return_periods] for d in boot_fits]
```

Line 2

Resample `length(y)` values with replacement, refit, and keep the fitted distribution.

```
lower = [quantile([c[k] for c in boot_curves], 0.025) for k in
eachindex(return_periods)]
upper = [quantile([c[k] for c in boot_curves], 0.975) for k in
eachindex(return_periods)]

fig = Figure(; size=(700, 400))
```

```
ax = Axis(fig[1, 1]; ylabel=L"\text{return level (ft)}")
band!(ax, return_periods, lower, upper; color=(colors[2], 0.25), label="95% bootstrap")
lines!(
    ax, return_periods, [quantile(fit_lmom, 1 - 1 / T) for T in return_periods];
    color=colors[2], linewidth=2, label="L-moments"
)
scatter!(ax, observed_periods, sorted_obs; color=:black, markersize=6,
label="observed")
return_period_axis!(ax)
axislegend(ax; position=:lt)
fig
```

Line 6

`band!` fills between the two quantile curves.

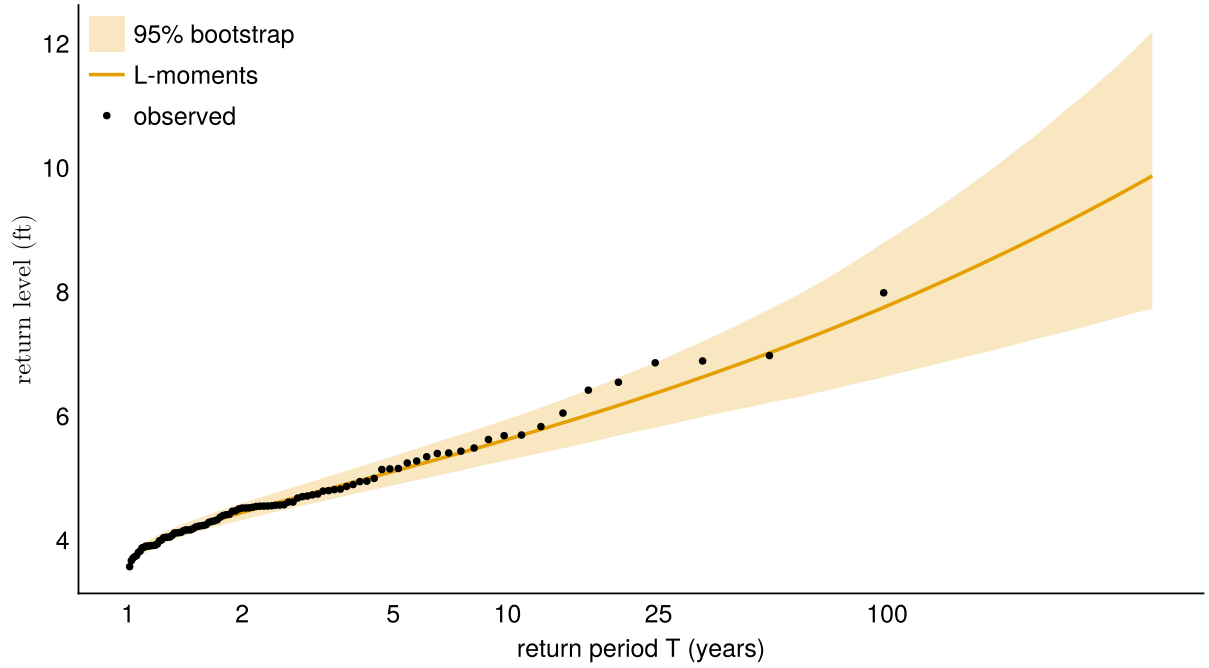


Figure 3: L-moment fit with a bootstrap 95% confidence band, against the observed record.

```
for T in (10, 50, 100, 500)
    levels = [quantile(d, 1 - 1 / T) for d in boot_fits]
    @printf("%3d-yr: %.2f [%.2f, %.2f] ft\n", T, quantile(fit_lmom, 1 - 1 / T),
        quantile(levels, 0.025), quantile(levels, 0.975))
end
```

```
 10-yr: 5.62 [5.29, 5.95] ft
 50-yr: 7.04 [6.23, 7.75] ft
```

```
100-yr: 7.77 [6.64, 8.83] ft
500-yr: 9.87 [7.72, 12.19] ft
```

The band widens with return period, because the far tail is determined by the few largest observations. Compare its width to the gap between the three fits in the previous figure: the disagreement between estimators is small next to the uncertainty within any one of them.

Bayesian estimation

The same model in Turing, with priors so wide they are meant to impose almost nothing. Priors this wide are often called uninformative.

```
@model function gev_vague(y)
    μ ~ Normal(0, 100)
    σ ~ truncated(Normal(0, 100); lower=0.01)
    ξ ~ Normal(0, 10)
    y .~ GeneralizedExtremeValue(μ, σ, ξ)
end
```

Line 3

Truncated so it stays positive.

Sampling the prior with `Prior()` ignores the data, so the draws show what the priors alone imply about water levels.

```
prior_df = DataFrame(sample(rng, gev_vague(y), Prior(), 1_000; progress=false))
```

Line 1

`Prior()` samples the prior rather than the posterior, which is what makes this a check on the prior alone.

```
function return_curves(df, periods; n_curves=200)
    curves = Vector{Vector{Float64}}()
    for i in 1:min(n_curves, nrow(df))
        df.σ[i] <= 0 && continue
        d = GeneralizedExtremeValue(df.μ[i], df.σ[i], df.ξ[i])
        curve = [quantile(d, 1 - 1 / T) for T in periods]
        all(isfinite, curve) && push!(curves, curve)
    end
    return curves
end

prior_curves = return_curves(prior_df, return_periods)
implied_500yr = [last(c) for c in prior_curves]
@printf("500-year level implied by the prior:\n")
@printf("  median %.3g ft, largest %.3g ft\n", median(implied_500yr),
maximum(implied_500yr))
```

Line 1

Turning each parameter draw into a return-level curve, which is the scale a coastal engineer can judge.

Line 7

A draw with ξ large enough gives infinite or overflowing quantiles, so those are dropped rather than plotted.

```
500-year level implied by the prior:
  median 4.65e+04 ft, largest 5.62e+85 ft
```

```
fig = Figure(; size=(700, 340))
ax = Axis(fig[1, 1]; ylabel=L"\text{return level (ft)}", title="vague prior")
for curve in prior_curves
    lines!(ax, return_periods, curve; color=(colors[1], 0.06))
end
ylims!(ax, -20, 60)
hlines!(ax, [maximum(y)]; color=:black, linestyle=:dash, label="largest observed")
return_period_axis!(ax)
axislegend(ax; position=:lt)
fig
```

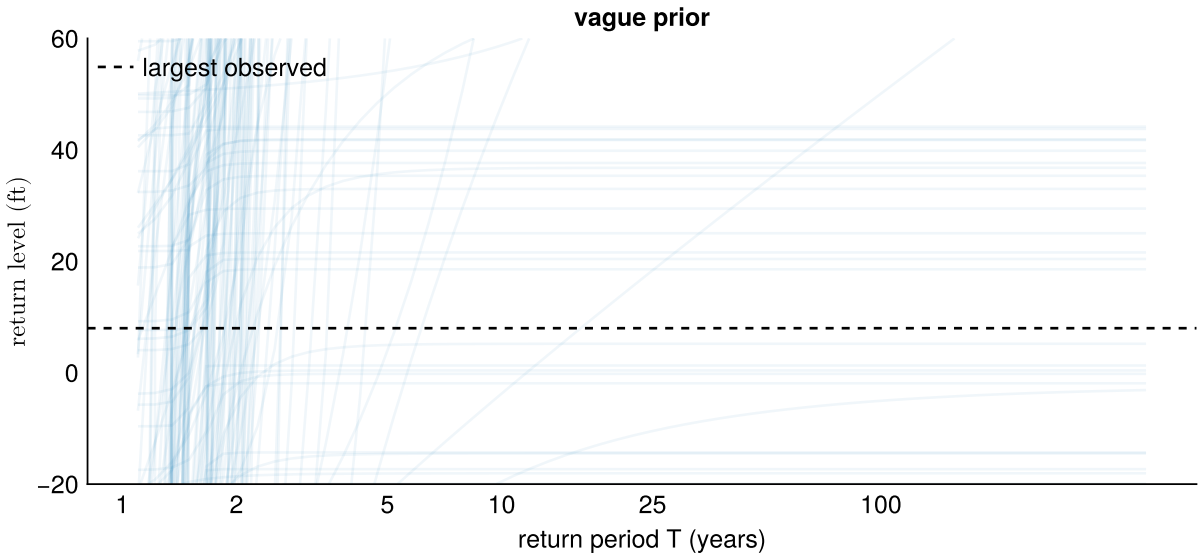


Figure 4: Return-level curves implied by the vague prior, before seeing any data. The vertical axis is clipped; many draws run far off the top.

These priors place substantial probability on 500-year water levels of several hundred feet and on negative water levels, neither of which any tide gauge has recorded. The prior is informative about return levels, and what it asserts is wrong.

Quantile priors

The three parameters are hard to reason about jointly, which is why independently reasonable priors on each one produced the curves above. Return levels are easier to judge, so specify the prior in terms of return levels instead. Choose a few return periods, state what you believe about the level at each, and let those statements constrain the parameters.

```
quantile_beliefs = [
    (period=2, belief=Normal(5.0, 1.0)),
    (period=100, belief=Normal(8.0, 2.5)),
    (period=500, belief=Normal(9.5, 4.0)),
]

@model function gev_quantile(y, beliefs)
    μ ~ Normal(5, 3)
    σ ~ truncated(Normal(0, 2); lower=0.01)
    ξ ~ truncated(Normal(0, 0.2); lower=-0.5, upper=0.5)
    d = GeneralizedExtremeValue(μ, σ, ξ)
    for b in beliefs
        Turing.@addlogprob! logpdf(b.belief, quantile(d, 1 - 1 / b.period))
    end
    for yᵢ in y
        Turing.@addlogprob! my_gev_logpdf(yᵢ, μ, σ, ξ)
    end
end
```

Line 1

Each entry is a belief about one return level: the 2-year level is 5 ft with standard deviation 1 ft, the 100-year level is 8 ft with standard deviation 2.5 ft, and the 500-year level is 9.5 ft with standard deviation 4 ft. **These numbers are for Sewells Point.** A gauge with a different tidal range needs different ones.

Line 10

Bounding the shape rules out the distributions with no finite variance. This is a strong assumption; revisit it if the posterior for ξ concentrates near a bound.

Line 13

Each belief contributes its log-density evaluated at the return level the current parameters imply, so parameter sets implying return levels you called implausible receive lower posterior density.

```
quantile_prior_df = DataFrame(
    sample(rng, gev_quantile(y, quantile_beliefs), Prior(), 1_000; progress=false)
)
quantile_curves = return_curves(quantile_prior_df, return_periods)

fig = Figure(; size=(700, 340))
ax = Axis(fig[1, 1]; ylabel=L"\text{return level (ft)}", title="quantile prior")
for curve in quantile_curves
```

```
    lines!(ax, return_periods, curve; color=(colors[1], 0.06))
end
hlines!(ax, [maximum(y)]; color=:black, linestyle=:dash, label="largest observed")
return_period_axis!(ax)
axislegend(ax; position=:lt)
fig
```

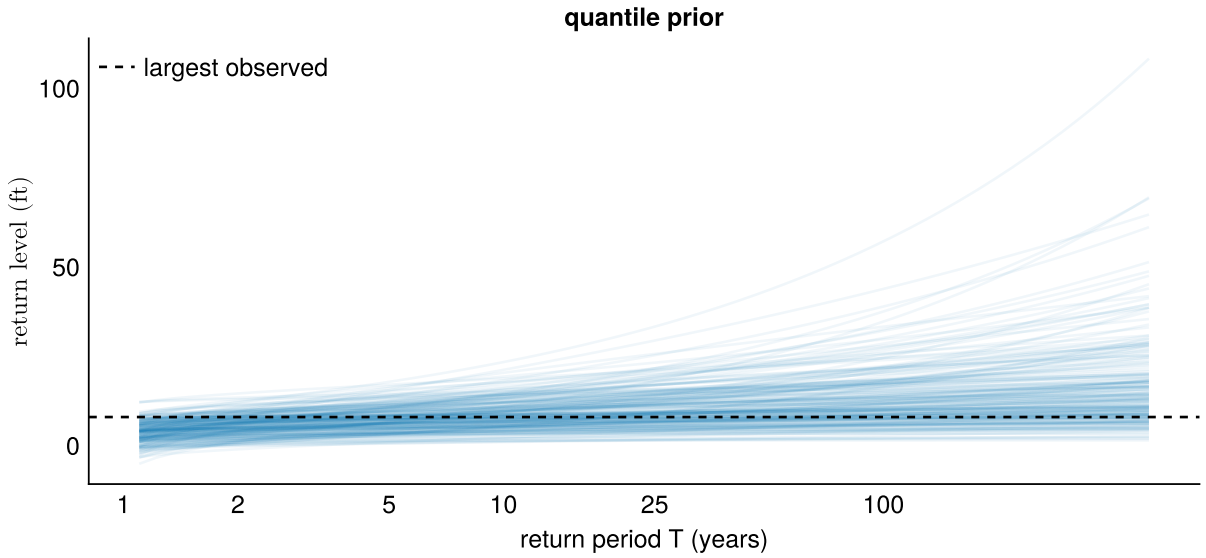


Figure 5: Return-level curves implied by the quantile prior. Compare the vertical scale to the previous figure.

The implied return levels are now physically plausible, and they still span a wide range. A prior predictive check that produces a narrow band means the prior determines the posterior.

Posterior fits and convergence diagnostics

```
chain_vague = sample(
    rng, gev_vague(y), NUTS(0.99), MCMCThreads(), 1_000, 4; progress=false
)
chain_quantile = sample(
    rng, gev_quantile(y, quantile_beliefs), NUTS(0.99), MCMCThreads(), 1_000, 4;
    progress=false,
)
```

Line 2

Four chains of 1,000 draws, run from different starting points. $\hat{R}$ compares chains against each other, so it needs more than one.

Line 5

Target acceptance rate, raised from the default of 0.65. The GEV support depends on the parameters, so the posterior density falls to zero along a boundary that moves with μ, σ and ξ. At the default step size NUTS crosses that boundary and reports a divergence. A higher target forces smaller steps, which costs a few seconds of runtime. Models whose support depends on their parameters usually need this; most models do not.

Check the chains before using them.

```
function check_chain(chain, label)
    stats = DataFrame(summarystats(chain))
    divergences = sum(chain[:numerical_error])
    @printf(
        "%-15s worst R̂ = %.3f, smallest ESS = %5.0f, divergences = %d\n",
        label, maximum(stats.rhat), minimum(stats.ess_bulk), divergences
    )
    return nothing
end

check_chain(chain_vague, "vague prior")
check_chain(chain_quantile, "quantile prior")
```

Line 1

Turing records a flag per draw; summing gives the count.

```
vague prior     worst R̂ = 1.003, smallest ESS =  1587, divergences = 0
quantile prior  worst R̂ = 1.003, smallest ESS =  1642, divergences = 0
```

> **i** Note
>
> **What the numbers mean.** $\hat{R}$ compares the variation within each chain to the variation between chains; at convergence it approaches 1, and anything above 1.01 means the chains have not mixed. The effective sample size is how many independent draws the correlated ones are worth, so 4,000 draws with an ESS of 200 carry the information of 200. Divergences should be as close to zero as possible.

```
fig = Figure(; size=(700, 400))
ax = Axis(fig[1, 1]; ylabel=L"\text{return level (ft)}")
for (chain, colour, label) in
     ((chain_vague, colors[1], "vague prior"), (chain_quantile, colors[2], "quantile
prior"))
    curves = return_curves(DataFrame(chain), return_periods; n_curves=2_000)
    lo = [quantile([c[k] for c in curves], 0.025) for k in eachindex(return_periods)]
    hi = [quantile([c[k] for c in curves], 0.975) for k in eachindex(return_periods)]
    med = [median(c[k] for c in curves) for k in eachindex(return_periods)]
```

```
    band!(ax, return_periods, lo, hi; color=(colour, 0.2))
    lines!(ax, return_periods, med; color=colour, linewidth=2.5, label=label)
end
scatter!(ax,     observed_periods,     sorted_obs;     color=:black,     markersize=6,
label="observed")
return_period_axis!(ax)
axislegend(ax; position=:lt, framevisible=true)
fig
```

Line 6

The 2.5th and 97.5th percentiles across posterior draws at each return period, which bound the 95% credible interval.

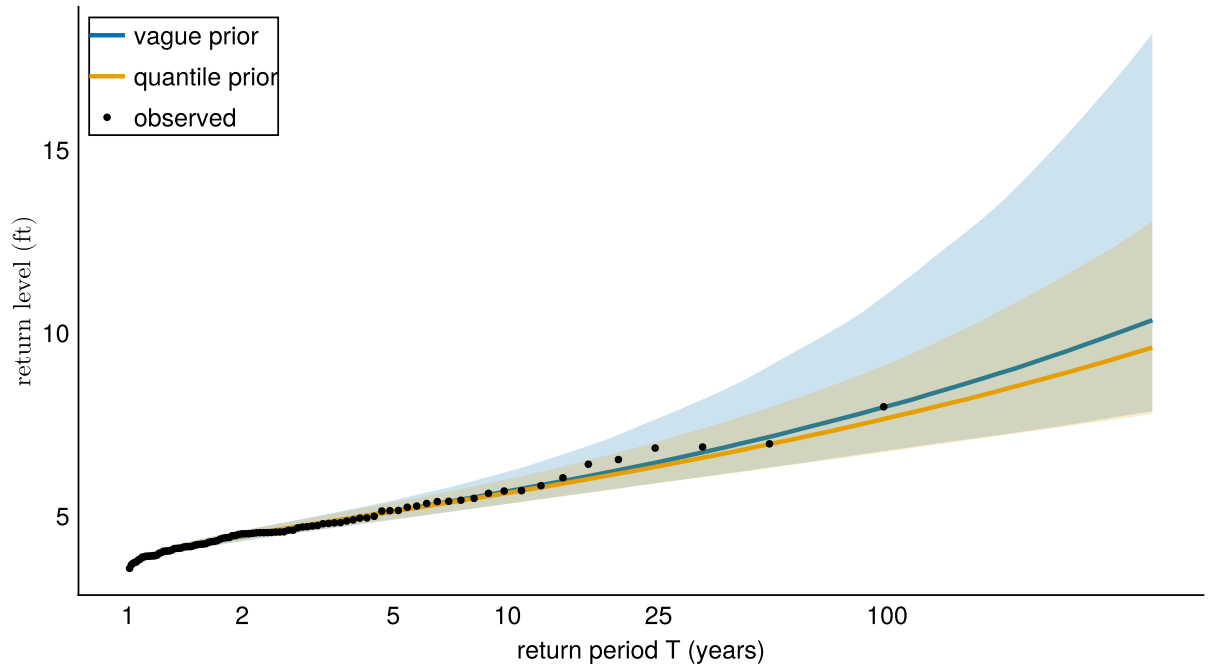


Figure 6: Posterior median return level and 95% credible interval under each prior. Shaded bands are the credible intervals; the observed record is plotted at Weibull plotting positions.

The two posteriors differ most at long return periods, where the data constrain the tail weakly.

```
function return_level_summary(chain, period)
    df = DataFrame(chain)
    levels = [
        quantile(GeneralizedExtremeValue(df.μ[i], df.σ[i], df.ξ[i]), 1 - 1 / period)
        for i in 1:nrow(df)
    ]
    return median(levels), quantile(levels, 0.025), quantile(levels, 0.975)
end
```

```
for period in (50, 100, 500)
    vm, vl, vu = return_level_summary(chain_vague, period)
    qm, ql, qu = return_level_summary(chain_quantile, period)
    @printf(
        "%3d-yr   vague %5.2f [%5.2f, %6.2f]   quantile %5.2f [%5.2f, %6.2f]\n",
        period, vm, vl, vu, qm, ql, qu
    )
end
```

Line 1

Every posterior draw is a full parameter vector, so every draw gives a return level. The spread of those levels is the credible interval, and it accounts for uncertainty in all three parameters at once.

```
 50-yr   vague  7.19 [ 6.36,   9.05]   quantile  6.98 [ 6.31,   7.99]
100-yr   vague  8.01 [ 6.81,  10.99]   quantile  7.67 [ 6.75,   9.17]
500-yr   vague 10.37 [ 7.88,  17.93]   quantile  9.60 [ 7.79,  13.10]
```

Exercises

You are expected to use an LLM for the code. Write your answers in the **Your answer** block under each exercise, replacing the italic placeholder. Grading is on the answers.

Change the gauge

Pick a different NOAA tide gauge from the station browser and change `station` in the data cell near the top. Choose one with at least 50 years of record. Then run the whole notebook from the top.

Report what broke and what you did about it. Work through the notebook in order and say, at each stage, whether the output is still trustworthy: did the hand-written optimizer converge, do the two fits still agree, is $\hat{R}$ still close to 1, are there divergences.

The quantile prior's numbers were chosen for Sewells Point. A gauge with a different tidal range, meaning the difference between high and low tide, needs different ones. Sewells Point has a tidal range of about 2.5 ft and Boston about 9.5 ft, so Boston's annual maxima exceed the levels this prior calls plausible. Look at your gauge's record before choosing new numbers, and say what you changed them to and why.

Your answer.

Answer here

The shape parameter

Compare the maximum likelihood, L-moment, and hand-optimized estimates on your gauge. Report all three.

They should agree closely on μ and σ and less closely on ξ. Explain why ξ is the least well determined, and say which of the three you would report to a client and why.

Your answer.

Answer here

Sensitivity to the prior

Compare the 500-year return level and its credible interval under the two priors on your gauge.

The vague prior is sometimes described as letting the data speak for themselves. Using your prior predictive figures and your posterior intervals, say whether that description holds here.

Your answer.

Answer here

Reading a failed diagnostic

Suppose a colleague sends you this notebook, run on their own gauge, with $\hat{R} = 1.08$ on ξ and 47 divergences, and asks whether the 100-year level in the summary is usable.

Write your reply. Say what the two numbers mean, whether the estimate can be used, and what you would ask them to try.

Your answer.

Answer here

Reporting a design value

Report one 100-year return level for your gauge, with one interval, as you would for a report a city will act on.

State the number and say what it rests on: which estimator, which detrending choice, which prior, and what you would want before trusting it further.

Your answer.

Answer here

Turning it in

This lab is due on Canvas one week after the Friday session that starts it.

1. Check that every **Your answer** block is filled in.
2. Render the notebook to a PDF. In the terminal, from the lab folder, run:

   ```
   quarto render index.qmd --to typst
   ```

   This writes `index.pdf` beside the notebook. It runs the whole notebook from the top in a fresh process, so it also confirms your work does not depend on leftover REPL state.
3. Commit and push your work, with GitHub Desktop or by asking Claude to commit and push.
4. Upload `index.pdf` to the Lab 4 assignment on Canvas.

5. Paste the link to your repository in the same Canvas submission.

If the render fails and you cannot fix it, upload what you have and say where it broke. A lab that does not render is worth more to me than a lab you did not turn in.

Note what broke, what Claude got wrong, and what you had to fix by hand, and bring it to the next session.

Bibliography